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Phishing and scam detection using AI

Empowering Users to Spot Phishing and Scam Messages Early: Advances in AI Detection, Explainability, and User-Centric Design

Quick Reference

Key Findings Table

Theme	Key Insights	Supporting Citations
Hybrid Detection & Explainability	Combining deep learning, NLP, LLMs, and XAI (e.g., SHAP, LIME) achieves >99% accuracy and user-friendly explanations.	1 2 3
Adaptability to Evolving Threats	Hybrid models with RL and retrieval-augmented generation adapt to zero-day and AI-generated phishing.	4 5 6 7
User-Centric Explainability & Trust	Simple, actionable explanations increase user trust and adoption; visual/textual cues are effective.	2 8 9 10
Computational Trade-offs	Model compression, hardware acceleration, and hybrid edge-cloud strategies optimize LLM deployment.	11 12 13
Long-term User Education Gap	Sustained behavioral change and robust explainability under adversarial attacks remain research gaps.	14 15 16 17

Direct Answer

To help people spot phishing and scam messages early, a promising strategy is to use a hybrid AI detection system that combines deep learning, NLP, and large language models with explainable AI approaches. This system evaluates message content and embedded link features, and then translates risk assessments into simple, actionable language (e.g., "This message has urgent language and an unverified link") to improve user understanding and trust 1 2 3.

Study Scope

- **Time Period:** Recent advances (2020–2024)
- **Disciplines:** Computer Science, Cybersecurity, Human-Computer Interaction, Psychology
- **Methods:** Meta-analysis, empirical studies, systematic reviews, user studies, simulation, and real-world deployment evaluations

Assumptions & Limitations

- Assumes access to diverse, representative datasets for phishing detection.
- Most studies focus on English-language phishing; multilingual and cross-platform detection is emerging.
- Explainability frameworks are not yet standardized; adversarial robustness varies.
- Long-term user education and behavioral change are not fully addressed by current solutions.

Suggested Further Research

- Standardizing explainability frameworks resilient to adversarial manipulation.
- Integrating adaptive, real-time phishing training with detection tools for sustained user vigilance.
- Expanding multilingual and cross-platform detection and explanation capabilities.
- Evaluating long-term effectiveness of gamified and personalized training in diverse organizational contexts.

1. Introduction

1.1 Background and Motivation

Phishing and scam messages are a persistent and evolving threat, exploiting human trust and technological vulnerabilities across email, SMS, social media, and websites. Attackers increasingly use AI-generated content, making detection harder for both users and traditional security systems. Despite advances in technical defenses, users remain susceptible due to limited awareness, cognitive overload, and the sophistication of modern phishing tactics [1] [14] [18].

Traditional detection methods and user education programs often fail to keep pace with evolving threats, especially zero-day and AI-generated attacks. Users struggle to identify scams early, and existing tools rarely provide clear, actionable explanations that foster trust and informed decision-making. There is a pressing need for AI-driven, explainable, and user-friendly solutions that empower users to spot phishing and scam messages before harm occurs [1] [14] [18].

1.2 Scope of the Report

This report synthesizes advances in AI-based phishing detection, focusing on:

- Hybrid models (deep learning, NLP, LLMs) for early detection.
- Explainability approaches that translate risk into simple language.
- User interaction, trust, and adoption factors.
- Gaps in current explainability and user education.
- Evidence-based strategies for long-term phishing awareness and behavioral change [1] [2] [9].

2. Theoretical Frameworks

2.1 Hybrid AI Detection and Explainability

Hybrid AI models combine deep learning (CNN, LSTM, TCN), NLP, and LLMs with explainable AI (XAI) techniques (SHAP, LIME, attention) to achieve high detection accuracy and user-friendly explanations. This dual capability is essential for real-time detection and user education [1] [2] [3].

Conceptual Model (No diagram provided)

- **Detection Layer:** Multi-modal analysis (text, URLs, images, metadata).
- **Explainability Layer:** XAI methods highlight key phishing indicators.
- **User Interface Layer:** Translates technical risk into simple, actionable language.

2.2 Adaptability to Evolving Threats

Emerging hybrid models integrate reinforcement learning (RL) and retrieval-augmented generation (RAG) to handle zero-day and AI-generated phishing attacks. These adaptive approaches continuously learn from new strategies, improving resilience and supporting dynamic, real-time responses [4] [5] [6] [7].

2.3 User-Centric Explainability and Trust

Explainability approaches that translate AI risk assessments into simple, actionable language increase user trust and facilitate better decision-making. Tools that incorporate clear textual explanations and visualizations enable non-technical users to understand key phishing indicators, enhancing system adoption [2] [8] [9] [10].

2.4 Computational Trade-offs

Deploying sophisticated LLM-based detection methods in resource-constrained environments requires balancing detection accuracy, latency, and computational overhead. Model compression, hardware acceleration, and hybrid edge-cloud strategies optimize performance while keeping the system lightweight [11] [12] [13].

3. Methods & Data Transparency

3.1 AI Techniques for Early Detection

- **Deep Learning:** CNN, LSTM, TCN models achieve >99% accuracy in complex phishing detection scenarios [1].
- **LLMs:** GPT-2, Llama-3, ChatGPT-4, Gemini excel in phishing email classification, leveraging NLP for semantic analysis [1] [19].
- **Traditional ML:** Random Forest, XGBoost perform well on structured datasets but lack semantic adaptability [1] [20] [21].
- **Ensemble & Hybrid Models:** AdaBoost, Bagging, Stacking, and pipelines combining ML, DL, NLP, and LLMs improve accuracy and robustness [22] [23].
- **Multi-modal Detection:** Integrates text, URLs, images, and metadata for cross-platform detection [24] [25] [26].

- **Reinforcement Learning & RAG:** RL (DQN) and RAG techniques adapt to zero-day and AI-generated phishing [4] [27].
- **Handling Imbalanced/Adversarial Data:** SMOTE, cost-sensitive learning, class-wise flooding, and adversarial training improve robustness [28] [29] [30].

3.2 Explainability Approaches

- **SHAP & LIME:** Provide global and local feature importance, enabling interpretable risk explanations [2] [31].
- **Attention Mechanisms:** Highlight relevant phishing indicators without overwhelming users [32] [33] [34].
- **LLMs for Explanation:** Generate context-aware, user-friendly explanations, improving trust and comprehension [35] [36] [37].
- **Human-Readable Frameworks (HRIT):** Transform technical features into semantically meaningful indicators without retraining detection models [38].
- **Multi-modal Explanation Integration:** Unifies outputs from text, URL, and image analysis into coherent interfaces [25] [39] [40].

3.3 User Studies & Real-world Deployment

- **User Trust & Interaction:** Real-world studies show that clear explanations and feedback improve trust and adoption [9] [41] [42] [43].
- **Demographic & Psychological Factors:** Age, gender, education, personality, and digital literacy affect susceptibility and trust [44] [45] [46].
- **Design Principles:** Balancing user autonomy with AI collaboration, explainability, and adaptive interfaces is key [8] [16] [47] [48].
- **Social Influence & Motivation:** Social factors and hedonic motivation drive long-term engagement [49] [50] [51].

4. Critical Analysis of Findings

4.1 AI Techniques for Early Phishing and Scam Detection

Deep Learning, LLMs, and Hybrid Models

- Deep learning models (CNN, LSTM, TCN) consistently achieve high accuracy (>99%) in phishing detection, especially with large or unstructured datasets [1].
- LLMs (GPT-2, Llama-3, ChatGPT-4) excel in semantic analysis, outperforming traditional ML in adaptability and contextual understanding [1] [19].
- Hybrid models combining ML, DL, NLP, and LLMs leverage complementary strengths, improving detection accuracy and robustness [22] [23].

- Ensemble methods (AdaBoost, Bagging, Stacking) further enhance performance across platforms [52].

Multi-Modal and Cross-Platform Detection

- Multi-modal frameworks fuse text, URLs, images, and metadata, enabling robust, real-time detection across emails, websites, SMS, and social media [24] [25] [26].
- Late fusion architectures and transformer-based models achieve high accuracy and low latency, suitable for real-world deployment [26] [53].

Adaptability to Zero-Day and AI-Generated Phishing

- RL and RAG techniques enable models to adapt to novel and AI-generated phishing attacks, outperforming static models in dynamic environments [4] [27].
- RL frameworks with human feedback improve detection accuracy and adaptability [54] [55].

Handling Imbalanced and Adversarial Datasets

- Techniques like SMOTE, cost-sensitive learning, class-wise flooding, and adversarial training improve detection performance on imbalanced and adversarial datasets [28] [29] [30].
- Feature-engineered representations enhance adversarial robustness and explainability [56].

Deployment and Computational Trade-offs

- LLM-based detection systems require significant computational resources; model compression, hardware acceleration, and hybrid edge-cloud strategies optimize latency and energy efficiency [11] [12] [13].
- Smaller open-source LLMs (7B–27B parameters) offer favorable trade-offs for resource-constrained environments [12].

Synthesis: Hybrid AI models, especially those integrating deep learning, NLP, LLMs, and XAI, provide highly accurate and robust phishing detection. Multi-modal and adaptive approaches are essential for handling evolving threats, while computational trade-offs must be managed for practical deployment [1] [4] [24].

4.2 Explainability: Making AI Risk Assessments User-Friendly

Translating AI Risk into Simple Language

- XAI techniques (SHAP, LIME, attention) highlight key phishing indicators (e.g., urgent language, suspicious URLs), improving user trust and actionable threat intelligence [2].
- LLMs generate understandable explanations aligned with recognizable phishing cues, balancing accuracy with operational trust [57] [58].
- Human-readable frameworks (HRIT) transform complex features into semantically meaningful indicators, optimized for LLM reasoning [38].

Optimizing Explanations for Relevance and Clarity

- Multimodal XAI approaches consolidate explanations, reducing user overwhelm [32].
- Feature selection frameworks and personalized explanation styles optimize relevance and interpretability [33] [59].
- Tri-class detection frameworks (Safe, Suspicious, Phishing) provide clearer, less overwhelming alerts [60].

Role of LLMs in Explanation Generation

- LLMs generate context-aware, user-friendly explanations, improving trust and comprehension [35] [36] [37].
- LLM-generated phishing warnings are perceived as more understandable and trustworthy than traditional browser warnings [58].

Human-Readable Frameworks and Semantic Feature Transformation

- HRIT enables interpretable reasoning about phishing intent without retraining detection models, enhancing explanation quality and auditability [38].

Integrating Multi-Modal Explanations

- Hybrid architectures and semantic-aware fusion frameworks organize multi-modal features for efficient, high-performance detection and explanation [25] [39] [40].
- LLM-based interpretation modules generate consolidated, human-readable explanations from raw scan results [40].

Synthesis: Explainability approaches that translate technical risk into simple, actionable language and visual cues significantly improve user trust and adoption. LLMs and human-readable frameworks are key to generating context-aware, user-friendly explanations [2] [35] [38].

4.3 User Interaction and Trust in AI-Based Phishing Detection

User Perceptions, Trust, and Real-World Interaction

- Real-time analysis with clear explanations enhances usability and alert clarity, increasing user acceptance [9].
- Trust is influenced by perceived competence, collaboration, and autonomy of the AI [41].
- Explanations help users calibrate trust, especially when AI accuracy is moderate [42].
- Feedback mechanisms improve trust calibration and user performance [43].

Demographic and Psychological Influences

- Age, gender, education, personality, and digital literacy affect trust, susceptibility, and adoption [44] [45] [46].
- Familiarity with AI and frequent technology use predict trust; older users are more skeptical [45].

Designing for Trust, Autonomy, and Collaboration

- Explainability and user interpretability are essential for balancing autonomy and AI collaboration [8] [47].
- Adaptive, explainable decision-making frameworks allow users to understand AI actions and maintain control [47].
- Real-time detection and response capabilities are critical for practical deployment [61].

Role of Social Influence and Motivation

- Social influence and hedonic motivation drive long-term engagement and sustained use [49] [50] [51].
- Trust moderates the relationship between gratifications and continued usage intention [49] [62].

Synthesis: User trust and adoption are shaped by clear explanations, feedback, demographic factors, and social influence. Design principles must balance user autonomy with effective AI collaboration, leveraging explainability and adaptive interfaces [9] [44] [47] [49].

4.4 Gaps in AI Explainability and User Education for Phishing Prevention

Limitations of Explainability and User Education

- Black-box AI models limit user understanding and trust; interpretable and actionable explanations are needed [14] [15] [16] [17].
- XAI faces challenges such as lack of standardized frameworks, methodological inconsistencies, and vulnerability to adversarial attacks [15] [63] [64] [65].
- Human judgment is constrained by slow behavioral adaptation and limited durability of awareness programs [16].

Adversarial Attacks and Explainability Vulnerabilities

- Adversarial attacks manipulate input features and exploit explainability mechanisms (SHAP, LIME) to evade detection [56] [66] [67].
- Defense mechanisms include adversarial training, input preprocessing, model regularization, and explainable defense architectures [67] [68] [69].

Usability and Adoption Challenges for Practitioners

- Balancing security effectiveness with usability is a core challenge; user-centered, seamless integration is needed [70] [71].
- Explainability and transparency are critical for user trust and adoption [10] [72] [73] [74] [75].
- Organizational and environmental factors influence adoption and effectiveness [76].

Beyond Traditional Training: Integrated Human-AI Frameworks

- Integrated human-AI decision frameworks combine adaptive AI detection with human behavioral insights, improving decision consistency and reducing errors [16].
- Agentic AI systems enable autonomous, context-aware detection and response [18] [47].
- Human-centered reinforcement learning with feedback improves adaptability and accuracy [77].

Synthesis: Despite advances, gaps remain in sustaining long-term user behavioral change and standardizing robust explainability methods that withstand adversarial challenges. Integrated human-AI frameworks and adaptive training are needed to address these limitations [14] [15] [16].

4.5 Advances in Phishing Awareness Training: Customization, Gamification, and Continuous Learning

Personalized and Adaptive Training Approaches

- Personalized training based on user proficiency improves detection skills and levels proficiency across groups [78] [79] [80].
- Mindfulness-based and overlearning strategies enhance long-term retention and reduce susceptibility [79] [81].
- Adaptive, behavior-driven frameworks with real-time feedback sustain accuracy and retention [80].

Gamification and Simulation-Based Education

- Gamification enhances engagement, motivation, and knowledge retention; simulation-based training improves awareness and reduces susceptibility [82] [83] [84].
- Adaptive gamification strategies tailored to individual learning styles and skill levels improve effectiveness [84] [85].

Evaluating Training Effectiveness in Organizations

- One-size-fits-all approaches neglect end-user preferences and organizational context; personalized guidelines are needed [86].
- Interactive and practical training methods (group discussions, role-playing) improve self-efficacy and support-seeking behaviors [87].
- Standardized evaluation metrics and bias-corrected methods are necessary for reliable assessment [88] [89].

Integrating Continuous, Adaptive Training with AI Detection

- Adaptive frameworks with continual learning, RL with human feedback, and online adaptation sustain detection accuracy over time [77] [90] [91] [92].
- Federated learning enables distributed, privacy-preserving models that adapt to emerging threats [93] [94].
- Multi-portal systems operationalize ongoing user engagement and awareness [95] [96].

Synthesis: Personalized, gamified, and adaptive training approaches, integrated with real-time AI detection, are essential for improving long-term retention and behavioral change in phishing awareness. Standardized

evaluation and continuous adaptation are key to sustaining user vigilance [78] [82] [86] [90].

5. Real-world Implications

5.1 Practical Deployment Strategies

- **Hybrid AI Systems:** Combining deep learning, NLP, LLMs, and XAI achieves high accuracy and user-friendly explanations, suitable for real-time deployment across platforms [1] [2] [3].
- **Multi-modal Detection:** Integrating text, URLs, images, and metadata enables robust detection in emails, websites, SMS, and social media [24] [25] [26].
- **Explainability Interfaces:** Translating risk assessments into simple language and visual cues improves user trust and adoption [2] [35] [38].
- **Adaptive Training:** Personalized, gamified, and continuous training sustains vigilance and reduces susceptibility [78] [82] [90].
- **Resource Optimization:** Model compression, hardware acceleration, and hybrid edge-cloud strategies optimize LLM deployment in resource-constrained environments [11] [12] [13].

5.2 User-Centric Design Principles

- **Clear Explanations:** Use simple, actionable language and visual cues to communicate risk.
- **Feedback Mechanisms:** Enable real-time user feedback to improve trust calibration and system adaptation.
- **Personalization:** Tailor training and explanations to user proficiency, demographic factors, and organizational context.
- **Social Influence:** Leverage social motivation and peer support to drive engagement and sustained use.
- **Privacy & Usability:** Ensure privacy-preserving, seamless integration with user workflows.

6. Future Research Directions

6.1 Standardizing Explainability Frameworks

- Develop robust, standardized XAI methods resilient to adversarial manipulation and suitable for diverse user groups [15] [63] [64] [65].

6.2 Integrating Adaptive, Real-Time Training

- Combine real-time AI detection with continuous, adaptive phishing training to sustain user vigilance and behavioral change [77] [90] [91] [92].

6.3 Expanding Multilingual and Cross-Platform Capabilities

- Enhance detection and explanation frameworks for multilingual and cross-platform phishing, addressing global and diverse user needs [97].

6.4 Evaluating Long-term Effectiveness

- Conduct longitudinal studies on the impact of gamified, personalized, and adaptive training in diverse organizational contexts [78] [82] [86].

7. Conclusion

7.1 Summary of Key Insights

Hybrid AI systems combining deep learning, NLP, LLMs, and XAI provide highly effective and accurate phishing detection while translating complex risk assessments into user-friendly, actionable insights. Adaptive techniques such as reinforcement learning are key for addressing evolving phishing strategies, and computational trade-offs must be managed carefully for real-world deployment. Despite significant advances, gaps remain in sustaining long-term user behavioral change and standardizing robust explainability methods that can withstand adversarial challenges [1] [2] [9] [14].

7.2 Recommendations for Research and Practice

- Adopt hybrid, multi-modal AI detection systems with integrated explainability.
- Design user interfaces that translate risk into simple, actionable language and visual cues.
- Implement personalized, gamified, and continuous training programs.
- Optimize deployment strategies for resource-constrained environments.
- Standardize explainability frameworks and evaluate long-term effectiveness in diverse contexts [1] [80] [98].

7.3 Emerging Trends and Open Challenges

- AI-generated phishing and adversarial robustness are growing threats.
- Holistic, adaptive, and explainable solutions are needed to empower users against evolving scams.
- Ongoing research should focus on integrating detection, explanation, and training for sustained user resilience [27] [92] [99].

Synthesis: The convergence of advanced AI detection, explainability, and user-centered design is transforming phishing prevention. Empowering users with clear, actionable insights and adaptive training is essential for mitigating the risks of increasingly sophisticated phishing and scam messages. Future research must address standardization, adversarial robustness, and sustained behavioral change to realize the full potential of AI-driven, user-friendly phishing defense.

References

1. A Systematic Review of Artificial Intelligence Techniques for Phishing Detection Aguirre, A., Salazar, L. Advances in Artificial Intelligence and Machine Learning, 2025

https://www.scopus.com/pages/publications/105013856311?origin=scopusAI

2. Demystifying Phishing Detection: Using Explainable AI to Identify Suspicious Email Features Gangrade, M., Sivasamy, S. Lecture Notes in Networks and Systems, 2026 https://www.scopus.com/pages/publications/105047816037?origin=scopusAI

3. An Outlook on Explainable AI Tools: SHAP and LIME Goel, A., Gupta, S., Gupta, P. Lecture Notes in Networks and Systems, 2026 https://www.scopus.com/pages/publications/105047339054?origin=scopusAI

4. Reinforcement learning for adaptive cyber defense against zero-day attacks Hu, Z., Chen, P., Zhu, M., Liu, P. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2019 https://www.scopus.com/pages/publications/85072066554?origin=scopusAI

5. Machine learning-based zero-day attack detection using deep reinforcement learning Hussain, S., Kalpana, V., Raghavendran, V., (...), Veeresha, R.K. Information Security Journal, 2026 https://www.scopus.com/pages/publications/105032150173?origin=scopusAI

6. Unveiling the Magic: Investigating Attention Distillation in Retrieval-Augmented Generation Li, Z., Zhang, H., Zhang, J. Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL 2024, 2024 https://www.scopus.com/pages/publications/85197752410?origin=scopusAI

7. Smishing risk assessment using RAG and its potential for pedagogical recontextualization for cybersecurity education Kalniņa, D., Norgaila, E., Daniela, L. Social Sciences and Humanities Open, 2026 https://www.scopus.com/pages/publications/105051206591?origin=scopusAI

8. Explainable AI-based Phishing Detection using TF-IDF-SVD Embeddings and XGBoost Classification Jeyaraj, J.R.A., Krishna, B.V., Reddy, B.N., (...), Naveen, V.V. 17th International Conference on Advances in Computing, Control, and Telecommunication Technologies, ACT 2026, 2026 https://www.scopus.com/pages/publications/105047272861?origin=scopusAI

9. PhishFind: A Machine Learning-Based System for Real-Time Phishing Detection Mendoza Vega, D.O., Diaz Mercado, D.O., Cardenas, E.J.E. Proceedings of 8th International Conference on Systems Engineering - Cybersecurity and AI: Building a reliable digital future, CIIS 2025, 2025 https://www.scopus.com/pages/publications/105025360349?origin=scopusAI

10. Explainable AI For Phishing Detection: Techniques, Challenges, and Experimental Validation Varsha, V.G., Thomas, P.A. RAICS - IEEE Recent Advances in Intelligent Computational Systems, 2025 https://www.scopus.com/pages/publications/105032735595?origin=scopusAI

11. Real-time and offline large language models on edge devices: a systematic review Dinçer, E., Kilimci, Z.H. PeerJ Computer Science, 2026 https://www.scopus.com/pages/publications/105038219306?origin=scopusAI

12. Benchmarking 21 Open-Source Large Language Models for Phishing Link Detection with Prompt Engineering Nasution, A.H., Monika, W., Onan, A., Murakami, Y. Information (Switzerland), 2025 https://www.scopus.com/pages/publications/105006642142?origin=scopusAI

13. Lightweight AI-Driven Model for Phishing Detection with Hash-Augmented Feature Embeddings Gaurav, A., Gupta, B.B., Wu, J., (...), Chui, K.T. GCCE 2025 - 2025 IEEE 14th Global Conference on Consumer Electronics, 2025 https://www.scopus.com/pages/publications/105031358517?origin=scopusAI

14. Security education against Phishing: A modest proposal for a Major Rethink Kirlappos, I., Sasse, M.A. IEEE Security and Privacy, 2012 https://www.scopus.com/pages/publications/84859726676?origin=scopusAI

15. Explainable Artificial Intelligence in Cyber Defense-A Review Kaur, G., Kaur, G., Choudhury, A., (...), Srivastava, A. 2025 7th International Symposium on Advanced Electrical and Communication Technologies, ISAECT 2025, 2025 https://www.scopus.com/pages/publications/105032703978?origin=scopusAI

16. Why educated users still click: a review on phishing and the limits of human judgment Gagniuc, P.A. Computer Science Review, 2026 https://www.scopus.com/pages/publications/105041375174?origin=scopusAI

17. Evaluating the Effectiveness of Security Awareness Programs and Developing Simulation-Based Methodologies to Enhance Cyber Attack Prevention Salih, M.E.M., Yahya Yousif, A.B., Babikir, S., (...), Arbab, A. 2025 International Conference on Decision Aid Sciences and Applications, DASA 2025, 2025 https://www.scopus.com/pages/publications/105040830082?origin=scopusAI

18. Phishing 2.0: exploring the capabilities and risks of agentic AI-enabled attacks Chandre, P., Bhujbal, P., Kerketta, R., (...), Bhosale, R. Frontiers in Computer Science, 2026 https://www.scopus.com/pages/publications/105040082029?origin=scopusAI

19. How effective are large language models in detecting phishing emails? Hua, J., Wang, P., Lutchkus, P. Issues in Information Systems, 2024 https://www.scopus.com/pages/publications/105003927799?origin=scopusAI

20. The State of the Art in AI-Based Phishing Detection: A Systematic Literature Review Alghenaim, M., Alkawsi, G., Barnhart, C.R. Studies in Computational Intelligence, 2025 https://www.scopus.com/pages/publications/85217941123?origin=scopusAI

21. An Evaluation and Comparison for Phishing Attack Detection using Machine Learning Approaches Sharma, A.K., Anushree, Rakesh, N., Verma, P.K. 2024 3rd International Conference on Power Electronics and IoT Applications in Renewable Energy and its Control, PARC 2024, 2024 https://www.scopus.com/pages/publications/85190991522?origin=scopusAI

22. A methodology combining NLP, Machine Learning, and LLM for the Detection of Phishing Emails Rosati, D., Aldini, A. CEUR Workshop Proceedings, 2026 https://www.scopus.com/pages/publications/105042500399?origin=scopusAI

23. Phishing Detection Using Machine Learning and Deep Learning Approaches Shah, P., Thakur, P.M., Yadav, R., Vishwakarma, D.K. Proceedings of the 6th International Conference on Inventive Research in Computing Applications, ICIRCA 2025, 2025 https://www.scopus.com/pages/publications/105013619056?origin=scopusAI

24. UEP-IQ: A Deployable Multi-modal Phishing Detection Framework Huynh, T.-T., Thanh, H.P., Le, G.H., (...), Trung, K.N. Communications in Computer and Information Science, 2026 https://www.scopus.com/pages/publications/105037888263?origin=scopusAI

25. Reality Check: Multi-Modal Phishing Detection Using Deep Learning Swetha, T., Ambika, L.G., Archana, E., (...), Priyanka, B.N. ICKECS 2026 - Proceedings of 4th IEEE International Conference on Knowledge Engineering and Communication Systems, 2026 https://www.scopus.com/pages/publications/105041875188?origin=scopusAI

26. Transformer-Based Multimodal Feature Fusion for Phishing Email Detection Chen, W., Xiao, X., Chen, G. Proceedings - 2025 10th International Conference on Cyber Security and Information Engineering, ICCSIE 2025, 2025 https://www.scopus.com/pages/publications/105034046460?origin=scopusAI

27. Augmenting phishing squatting detection with GANs Valentim, R., Drago, I., Trevisan, M., (...), Mellia, M. CoNEXT-SW 2021 - Proceedings of the 2021 CoNEXT Student Workshop 2021 - Part of CoNEXT 2021 the 17th International Conference on emerging Networking EXperiments and Technologies, 2021 https://www.scopus.com/pages/publications/85121607283?origin=scopusAI

28. Real-time detectors for digital and physical adversarial inputs to perception systems Kantaros, Y., Carpenter, T., Sridhar, K., (...), Weimer, J. ICCPS 2021 - Proceedings of the 2021 ACM/IEEE 12th International Conference on Cyber-Physical Systems (with CPS-IoT Week 2021), 2021 https://www.scopus.com/pages/publications/85104201187?origin=scopusAI

29. Mitigating Data Imbalance Challenges in Deep Learning Models for Intrusion Detection in Cloud Computing Mala, K., Prashanth Kumar, S.P., Praveen Kumar, K.C., Mohan Kumar, T.G. 3rd International

Conference on Data Science and Information System, ICDSIS 2025, 2025 https://www.scopus.com/pages/publications/105011825688?origin=scopusAI

30. Class-Wise Flooding Regularization for Imbalanced Image Classification Aizawa, H., Naito, Y., Fukuda, K. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105022716662?origin=scopusAI

31. AI-Based Phishing Email and Link Detection Framework Through Generative AI and Explainable ML Sathiyapriya, S., Subhasri, B., Deepa, D. Proceedings of the 4th International Conference on Intelligent Data Communication Technologies and Internet of Things, IDCioT 2026, 2026 https://www.scopus.com/pages/publications/105037002929?origin=scopusAI

32. A Multimodal Phishing Website Detection System Using Explainable Artificial Intelligence Technologies Vulfin, A., Sulavko, A., Vasiliev, V., (...), Samotuga, A. Machine Learning and Knowledge Extraction, 2026 https://www.scopus.com/pages/publications/105028645835?origin=scopusAI

33. Exploring the impact of explainable AI and cognitive capabilities on users' decisions Cau, F.M., Spano, L.D. User Modeling and User-Adapted Interaction, 2026 https://www.scopus.com/pages/publications/105024066666?origin=scopusAI

34. An Explainable Multi-Scale Residual Attention Network with XGBoost Ensemble for Phishing Website Detection Chaudhary, R., Saxena, R., Dillu, V. International Journal of Intelligent Engineering and Systems, 2026 https://www.scopus.com/pages/publications/105044862856?origin=scopusAI

35. PhishLLM: Generating Japanese Phishing Emails to Test Cross-Lingual Generalization in English-Trained Models Meléndez, R., Ptaszynski, M., Masui, F. CEUR Workshop Proceedings, 2026 https://www.scopus.com/pages/publications/105040734142?origin=scopusAI

36. Toward Human-Centered Explainability: Natural Language Explanations for Anomaly Detection Padín-Torrente, H., Carneiro-Diaz, V., Ortega-Fernandez, I. Information Systems Frontiers, 2026 https://www.scopus.com/pages/publications/105035464519?origin=scopusAI

37. Can You Walk Me Through It? Explainable SMS Phishing Detection using LLM-based Agents Wang, Y., Zhai, H., Wang, C., (...), Wang, G. Proceedings of the 21st Symposium on Usable Privacy and Security, SOUPS 2025, 2025 https://www.scopus.com/pages/publications/105021086209?origin=scopusAI

38. HRIT: A Human-Readable Framework for Phishing URL Detection using Large Language Models Yadav, T., Masum, M. 2026 IEEE 5th International Conference on AI in Cybersecurity, ICAIC 2026, 2026 https://www.scopus.com/pages/publications/105041770387?origin=scopusAI

39. LLM-Based Multimodal Feature Extraction and Hierarchical Fusion for Phishing Email Detection Yuan, X., Wang, J., Yan, T., Qi, F. Electronics (Switzerland), 2026 https://www.scopus.com/pages/publications/105028583626?origin=scopusAI

40. Sentrashield: Guarding your Digital Frontiers with Precision Thejaswini, S., Anjithkrishnan, K., Sajith, D.K., (...), Mohan, M. Proceedings of the 2026 IEEE Nexo-Tech International Conference on Advanced Technologies and Innovations, ICATI 2026, 2026 https://www.scopus.com/pages/publications/105046328240?origin=scopusAI

41. User Perceptions and Trust of Explainable Machine Learning Fake News Detectors Shin, J., Chan-Olmsted, S. International Journal of Communication, 2023 https://www.scopus.com/pages/publications/85177890168?origin=scopusAI

42. IMPACT OF EXPECTATION AND PERFORMANCE ON THE USER EXPERIENCE OF AI SYSTEMS Park, S., Kim, H.K., Song, Y., (...), Park, J. ICIC Express Letters, Part B: Applications, 2022 https://www.scopus.com/pages/publications/85123102771?origin=scopusAI

43. The description-experience gap in the effect of warning reliability on user trust and performance in a phishing-detection context Chen, J., Mishler, S., Hu, B., (...), Proctor, R.W. International Journal of Human

Computer Studies, 2018 https://www.scopus.com/pages/publications/85050359479?origin=scopusAI

44. Trust and Acceptance Challenges in the Adoption of AI Applications in Health Care: Quantitative Survey Analysis Kauttonen, J., Rousi, R., Alamäki, A. Journal of Medical Internet Research, 2025 https://www.scopus.com/pages/publications/105001296976?origin=scopusAI

45. Demographic influences on trust in artificial intelligence across cognitive domains: A statistical perspective Alasmari, A.A., Alruwaili, R.F., Alotaibi, R.F., (...), Asklany, S.A. PLOS ONE, 2025 https://www.scopus.com/pages/publications/105020818201?origin=scopusAI

46. Human factors in phishing: Understanding susceptibility and resilience Oner, U., Cetin, O., Savas, E. Computer Standards and Interfaces, 2025 https://www.scopus.com/pages/publications/105003376786?origin=scopusAI

47. LLM-Based Autonomous Agents for Phishing Detection, Triage, and Response: A Reference Architecture for Agentic Cyber Defense Bazi, I.E., Ezzine, J., Ghazoui, B. Agentic Intelligence and the Rise of Autonomous Cyber Warfare, 2026 https://www.scopus.com/pages/publications/105047280016?origin=scopusAI

48. The Synthesis of Deception: AI-Driven Visual Personification and Adaptive Phishing in Modern Social Engineering Axmadaliyev, A., Jumanova, Z., Yusupov, D., (...), Marisheva, L. International Conference of Young Specialists on Micro/Nanotechnologies and Electron Devices, EDM, 2026 https://www.scopus.com/pages/publications/105048083487?origin=scopusAI

49. Psychological drivers and barriers affecting students' sustained usage intention of AIGC tools in educational settings: an integrated perspective of Uses and Gratifications Theory and Perceived Value Theory Huang, Q., Xu, J., Zhang, R., Song, G. Frontiers in Psychology, 2026 https://www.scopus.com/pages/publications/105049687921?origin=scopusAI

50. A longitudinal model of continued acceptance of conversational artificial intelligence Ng, Y.-L. Information Technology and People, 2025 https://www.scopus.com/pages/publications/85191860974?origin=scopusAI

51. Decoding User Intentions Towards AI Chatbot Services Under the Impact of Social Influences Truong, D.T., Tran, T.D., Phuong, T.N., (...), Phu, N.T.C. Emerging Science Journal, 2025 https://www.scopus.com/pages/publications/105009957963?origin=scopusAI

52. An Investigation of AI-based Ensemble Methods for the Detection of Phishing Attacks Alsariera, Y.A., Alanazi, M.H., Said, Y., Allan, F. Engineering, Technology and Applied Science Research, 2024 https://www.scopus.com/pages/publications/85196099400?origin=scopusAI

53. An expert system for phishing detection: a multimodal hybrid framework integrating deep and shallow machine learning Preeti, Priti Neural Computing and Applications, 2026 https://www.scopus.com/pages/publications/105038989670?origin=scopusAI

54. Human in the loop reinforcement learning framework for adaptive phishing detection Khan, S. Discover Computing, 2025 https://www.scopus.com/pages/publications/105025099708?origin=scopusAI

55. AI-Driven Phishing Detection: Enhancing Cybersecurity with Reinforcement Learning Jabbar, H., Al-Janabi, S. Journal of Cybersecurity and Privacy, 2025 https://www.scopus.com/pages/publications/105009276758?origin=scopusAI

56. Cross-Domain Adversarial Robustness and Explainability Stability in Phishing Detection Systems Hassan, Md.R., Nabin, J.A., Toyeer-E-Ferdoush 2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure, PECCII 2026, 2026 https://www.scopus.com/pages/publications/105049664527?origin=scopusAI

57. Fine-Tuning LLMs for Operational Phishing Email Detection Piugie, A.F.T., Valois, M., Giguet, E., (...), Chauvat, P. Proceedings of the International Conference on Security and Cryptography, 2026 https://www.scopus.com/pages/publications/105048601157?origin=scopusAI

58. APOLLO: A GPT-based tool to detect phishing emails and generate explanations that warn users Desolda, G., Greco, F., Vigano, L. Proceedings of the ACM on Human-Computer Interaction, 2025 https://www.scopus.com/pages/publications/105009641926?origin=scopusAI

59. Hybridizing Explainable AI (XAI) for Intelligent Feature Extraction in Phishing Website Detection Alsakarnah, R., Masoud, M.Z., Ghababsheh, A. Electronics (Switzerland), 2026 https://www.scopus.com/pages/publications/105028657506?origin=scopusAI

60. An Explainable Quantum-Inspired QUBO Feature Selection Framework for Risk Aware Tri-Class Phishing URL Detection Rahman Khan, M.M., Chowdhury, A.P., Nur, F.N. 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence and Networking, QPAIN 2026, 2026 https://www.scopus.com/pages/publications/105042925485?origin=scopusAI

61. A Review of Innovative Strategies for Mitigating Last-Mile Aggregation Attacks Utilizing Artificial Intelligence Abbas, I.A., Hasan, T.M. ICCES 2026 - Conference Proceedings of 2026 the 2nd International Conference Computing and Emerging Sciences, 2026 https://www.scopus.com/pages/publications/105038053087?origin=scopusAI

62. The Artificial Intelligence Use Motivations Scale (AIUMS): Development, Validation, and their Role in Problematic Generative AI Use (PGAU) Akbari, M., Mollaali Farkhani, S., Golmohammadi, M., (...), Griffiths, M.D. International Journal of Mental Health and Addiction, 2026 https://www.scopus.com/pages/publications/105040547661?origin=scopusAI

63. Explainable Artificial Intelligence in CyberSecurity: A Survey Capuano, N., Fenza, G., Loia, V., Stanzione, C. IEEE Access, 2022 https://www.scopus.com/pages/publications/85137876104?origin=scopusAI

64. A Systematic Review of Explainable Machine Learning Approaches in Phishing URL Detection Patidar, M., Panthi, S., Zalawadia, K. IEEE International Conference on Innovate for Humanitarian: Tech Solutions for Global Challenge, ICIH 2025, 2025 https://www.scopus.com/pages/publications/105046435071?origin=scopusAI

65. Security Challenges and Safeguards in Explainable Artificial Intelligence Abdullah, A.S., Dhiman, S. Attacks and Defenses in Explainable Artificial Intelligence, 2026 https://www.scopus.com/pages/publications/105039408732?origin=scopusAI

66. Vulnerability Assessment for Deep Learning Based Phishing Detection System Ogawa, Y., Kimura, T., Cheng, J. 2021 IEEE International Conference on Consumer Electronics-Taiwan, ICCE-TW 2021, 2021 https://www.scopus.com/pages/publications/85123051537?origin=scopusAI

67. Designing Explainable Defenses Against Sophisticated Adversarial Attacks Lakhanpal, S., Devi, K.V.R., Aravinda, K.A., (...), Kumar, A. Proceedings - 2024 13th IEEE International Conference on Communication Systems and Network Technologies, CSNT 2024, 2024 https://www.scopus.com/pages/publications/85196767670?origin=scopusAI

68. Adversarial Attacks on Neural Networks Pranjal, Vaishnavi, Divyansh, (...), Agarwal, A.K. Lecture Notes in Networks and Systems, 2024 https://www.scopus.com/pages/publications/85200495390?origin=scopusAI

69. Defense mechanisms against adversarial attacks: Strengthening AI security in cybersecurity applications Tariq, M.U. Challenges and Solutions for Cybersecurity and Adversarial Machine Learning, 2025 https://www.scopus.com/pages/publications/105011410320?origin=scopusAI

70. Driving 2FA adoption at scale: Optimizing two-factor authentication notification design patterns Golla, M., Ho, G., Lohmus, M., (...), Redmiles, E.M. Proceedings of the 30th USENIX Security Symposium, 2021 https://www.scopus.com/pages/publications/85114514749?origin=scopusAI

71. Enhancing Cybersecurity Through User Experience (UX) Design: Bridging Usability and Protection Wiberg, C. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105028364080?origin=scopusAI

72. Interpretable Artificial Intelligence for Cyber Defense: Advances, Limitations, and Future Opportunities Sammy, F., Thanammal Indu, V., Saranya, R., (...), Ezhilvendan, M. Proceedings of the 2026 12th International Conference on Communication and Signal Processing, ICCSP 2026, 2026 https://www.scopus.com/pages/publications/105042287966?origin=scopusAI

73. Mitigating Cyber Threats: Machine Learning and Explainable AI for Phishing Detection Akhtar, H.M.U., Nauman, M., Hameed, N.A.M., (...), Tareen, M.Z. VFAST Transactions on Software Engineering, 2025 https://www.scopus.com/pages/publications/105018782845?origin=scopusAI

74. Explaining the Suspicion: Design of an XAI-Based User-Focused Anti-Phishing Measure Kluge, K., Eckhardt, R. Lecture Notes in Information Systems and Organisation, 2021 https://www.scopus.com/pages/publications/85118158942?origin=scopusAI

75. AI-Enhanced Cybersecurity Operations System for Real-Time Phishing Detection and User Education Deji-Akinpelu, O., Olubunmi, C.O., Philip-Neris Nikatsekpe, T. Proceedings of 2026 1st International Conference on Intelligent Computing and Automation for Sustainable Solutions, ICASS 2026, 2026 https://www.scopus.com/pages/publications/105042707332?origin=scopusAI

76. How to Successfully Implement Phishing Awareness Training in Organizations: A Technology Adoption Perspective Geppert, T., Dudas, T., Zimmermann, S., (...), Ebert, N. Proceedings of the Annual Hawaii International Conference on System Sciences, 2025 https://www.scopus.com/pages/publications/105005161336?origin=scopusAI

77. Adaptive Phishing Detection and Mitigation System Using Huawei Mind Reinforcement Learning with Human Feedback † Hael, J.I.B., Ortiz, M.D.S., Padilla, D.A. Engineering Proceedings, 2026 https://www.scopus.com/pages/publications/105051806746?origin=scopusAI

78. It's a Match - Enhancing the Fit between Users and Phishing Training through Personalisation Schöni, L., Roch, N., Sievers, H., (...), Zimmermann, V. Conference on Human Factors in Computing Systems - Proceedings , 2025 https://www.scopus.com/pages/publications/105005712041?origin=scopusAI

79. Learning not to take the bait: a longitudinal examination of digital training methods and overlearning on phishing susceptibility Nguyen, C., Jensen, M., Day, E. European Journal of Information Systems, 2023 https://www.scopus.com/pages/publications/85108252234?origin=scopusAI

80. A Data-Driven Adaptive Cybersecurity Training Framework with Behavioral Validation Adamu, Y.A., Chaudhry, S.A., Zakaria, K., Yahya, K. International Journal of Networked and Distributed Computing, 2026 https://www.scopus.com/pages/publications/105046939345?origin=scopusAI

81. Training to Mitigate Phishing Attacks Using Mindfulness Techniques Jensen, M.L., Dinger, M., Wright, R.T., Thatcher, J.B. Journal of Management Information Systems, 2017 https://www.scopus.com/pages/publications/85027685681?origin=scopusAI

82. Cybersecurity Education Gamification: A Current Review and Research Agenda Salama, R., Al-Turjman, F. AI and Machine Learning for Mechanical and Electrical Engineering, 2025 https://www.scopus.com/pages/publications/105020130645?origin=scopusAI

83. Evaluating the Effectiveness of Customised Phishing Simulations in Enhancing Cybersecurity Awareness at a University in Thailand Naparat, D., Muensaen, O., Chaikaew, W. Lecture Notes on Data Engineering and Communications Technologies, 2025 https://www.scopus.com/pages/publications/105002282802?origin=scopusAI

84. Adaptive Gamification Strategies for Different Learning Styles in Cybersecurity Sindiramutty, S.R., Riskhan, B., Humayun, M. Gamification Learning Framework for Cybersecurity Education, 2025 https://www.scopus.com/pages/publications/105017943014?origin=scopusAI

85. Enhancing Cyber Security Education and Training through Gamification Diakoumakos, I. ACM International Conference Proceeding Series, 2023 https://www.scopus.com/pages/publications/85174513167?origin=scopusAI

86. Personalized Guidelines for Design, Implementation and Evaluation of Anti-Phishing Interventions Sarker, O., Haggag, S., Jayatilaka, A., Liu, C. International Symposium on Empirical Software Engineering and Measurement, 2023 https://www.scopus.com/pages/publications/85178657706?origin=scopusAI

87. The Effects of Group Discussion and Role-playing Training on Self-efficacy, Support-seeking, and Reporting Phishing Emails: Evidence from a Mixed-design Experiment Chen, X., Sacre, M., Lenzini, G., (...), Sergeeva, A. Conference on Human Factors in Computing Systems - Proceedings, 2024 https://www.scopus.com/pages/publications/85194819053?origin=scopusAI

88. Scaling the Phish: Advancing the NIST Phish Scale Barrientos, F., Jacobs, J., Dawkins, S. Communications in Computer and Information Science, 2021 https://www.scopus.com/pages/publications/85112059320?origin=scopusAI

89. Measuring the effectiveness of embedded phishing exercises Siadati, H., Palka, S., Siegel, A., McCoy, D. 10th USENIX Workshop on Cyber Security Experimentation and Test, CSET 2017, co-located with USENIX Security 2017, 2017 https://www.scopus.com/pages/publications/85084160985?origin=scopusAI

90. Adaptive transfer learning for robust phishing attack detection using recurrent layers: Enhancing cybersecurity through dynamic defense mechanisms Tayyeh, A.M., Hamad, A.A., Yaseen, R.B., (...), Abbas, O.A. Integrating Intelligent Control Systems With Sensor Technologies, 2025 https://www.scopus.com/pages/publications/105007490468?origin=scopusAI

91. Real-Time Incremental Transformer with Continual Learning for Adaptive Phishing Email Detection Suhaimee, M.N., Shakil, F., Khan, M.R.A.A., (...), Firdaus, S. 2025 IEEE 2nd International Conference on Computing, Applications and Systems, COMPAS 2025, 2025 https://www.scopus.com/pages/publications/105034658850?origin=scopusAI

92. AI-Driven Adaptive Phishing Mitigation in SOAR-Orchestrated Systems Sreelakshmi, M., Vasudevan, N.V., Aparna, P.V. 10th International Conference on Inventive Systems and Control, ICISC 2026 - Proceedings, 2026 https://www.scopus.com/pages/publications/105052020913?origin=scopusAI

93. Towards Trustworthy Email Phishing Detection: Integrating Multi-Modal Deep Learning, Federated Learning, and Explainable AI Adeyemo, A.A., Emuoyibofarhe, O.N., Abiodun, A.O., (...), Ajagbe, S.A. Proceedings of Machine Learning Research, 2026 https://www.scopus.com/pages/publications/105045819987?origin=scopusAI

94. Exploring the Efficacy of Federated-Continual Learning Nodes with Attention-Based Classifier for Robust Web Phishing Detection: An Empirical Investigation Jesher Joshua, M., Adhithya, R., Sree Dananjay, S., Revathi, M. International Conference on Advancements in Power, Communication and Intelligent Systems, APCI 2024, 2024 https://www.scopus.com/pages/publications/85201387376?origin=scopusAI

95. Phishlyze: A Hybrid AI Framework Combining Gradient Boosting And LLM Analysis For Advanced Phishing Detection Rajasekaran, M., Manjunadha Swamy, K., Sai Ganesh, T. 2026 8th International Conference on Information Technologies in Engineering Education, Inforino 2026 - Proceedings, 2026 https://www.scopus.com/pages/publications/105042954083?origin=scopusAI

96. Smart-Shield: A Hybrid ML-AI Framework for Advanced Phishing Detection using Gradient Boosting and LLM Analysis Kumar, P.V.S., Yadav, M.N., Reddy, E.C., Sathya, V. Proceedings of 5th International Conference on Pervasive Computing and Social Networking, ICPCSN 2025, 2025 https://www.scopus.com/pages/publications/105010213828?origin=scopusAI

97. Multilingual Email Phishing Attacks Detection using Open-Source Intelligence and Machine Learning An, P., Shafi, R., Mughogho, T., (...), Creutzburg, R. IS and T International Symposium on Electronic Imaging Science and Technology, 2026 https://www.scopus.com/pages/publications/105043728260?origin=scopusAI

98. Towards Explainable and Educational Phishing Detection: A Zero-Shot LLM Approach Zhang, L., Wu, Y. Proceedings of the IEEE International Conference on Big Data, BigData, 2025 https://www.scopus.com/pages/publications/105037875453?origin=scopusAI

99. Spam and Phishing Detection with Adversarial Attack Robustness Analysis Using Machine Learning Models Rączka, G., Plichta, A. Proceedings - European Council for Modelling and Simulation, ECMS, 2026 https://www.scopus.com/pages/publications/105042236606?origin=scopusAI